AMRITA SCHOOL OF ENGINEERING, BENGALURU

DEPARTMENT OF ELECTRONICS & COMMUNICATION

ENGINEERING

B. TECH. II SEMESTER AY 2024-2025

23ECE281 DIGITAL ELECTRONICS LAB

SYSTEM DESIGN REPORT

DIGITAL CLOCK

Submitted by

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COMMENTS (IF ANY):

AIM:
To design and implement a 12-hour digital clock system that accurately displays the time on common cathode 7-segment displays.

COMPONENTS REQUIRED ALONG WITH THEIR SPECIFICATIONS:

Sl. No.	Component	Specification	Count
1.	IC	CD4026BE	6
2.	IC	LS7411	1
3.	IC	NE555	1
4.	Resistors	470 ohms	8
5.	Resistor	1K ohm	1
6.	Potentiometer	100 K	1
7.	Capacitors	100 nf	1
8.	Capacitors	10 uf	1
9.	PCB		1
10.	Connecting Wires		-
11.	7 Segment Cathode Display		6
12.	Battery	1200 mah	1
13.	TP 4056 charging module		1
14.	(3.7 to 5) volts converter		1

THEORY: A digital clock is an electronic device that displays time using digital components like counters, logic gates and 7-segment displays. A digital clock works by continuously counting electronic pulses to keep track of the passing time and displaying this time on a digital display. Digital clocks use a series of counters to keep track of seconds, minutes, and hours.

IC 4026:
The IC 4026 is a decade counter with a built-in 7-segment display driver, counting from 0 to 9 with each clock pulse. It directly drives a 7-segment display, making it ideal for digital counters and clocks without needing extra decoding circuitry. Additionally, it has a carry-out pin to cascade multiple ICs for counting higher numbers, such as in multi-digit displays.
The circuit design has 6 counters in total, 2 ICs for seconds (units and tens place), 2 ICs for minutes (units and tens place) and 2 ICs for hours (units and tens place).

IC 7411 (3-Input AND Gate):
This IC contains three independent AND gates, each with three inputs. When all three inputs of any gate are HIGH, the output is HIGH; otherwise, the output is LOW. It can be used in a digital clock to handle reset logic by combining outputs from counters. When counting to 60 seconds or 60 minutes, the 7411 AND gate ensures all necessary conditions (e.g., 6 and 0) are met to reset the counters.

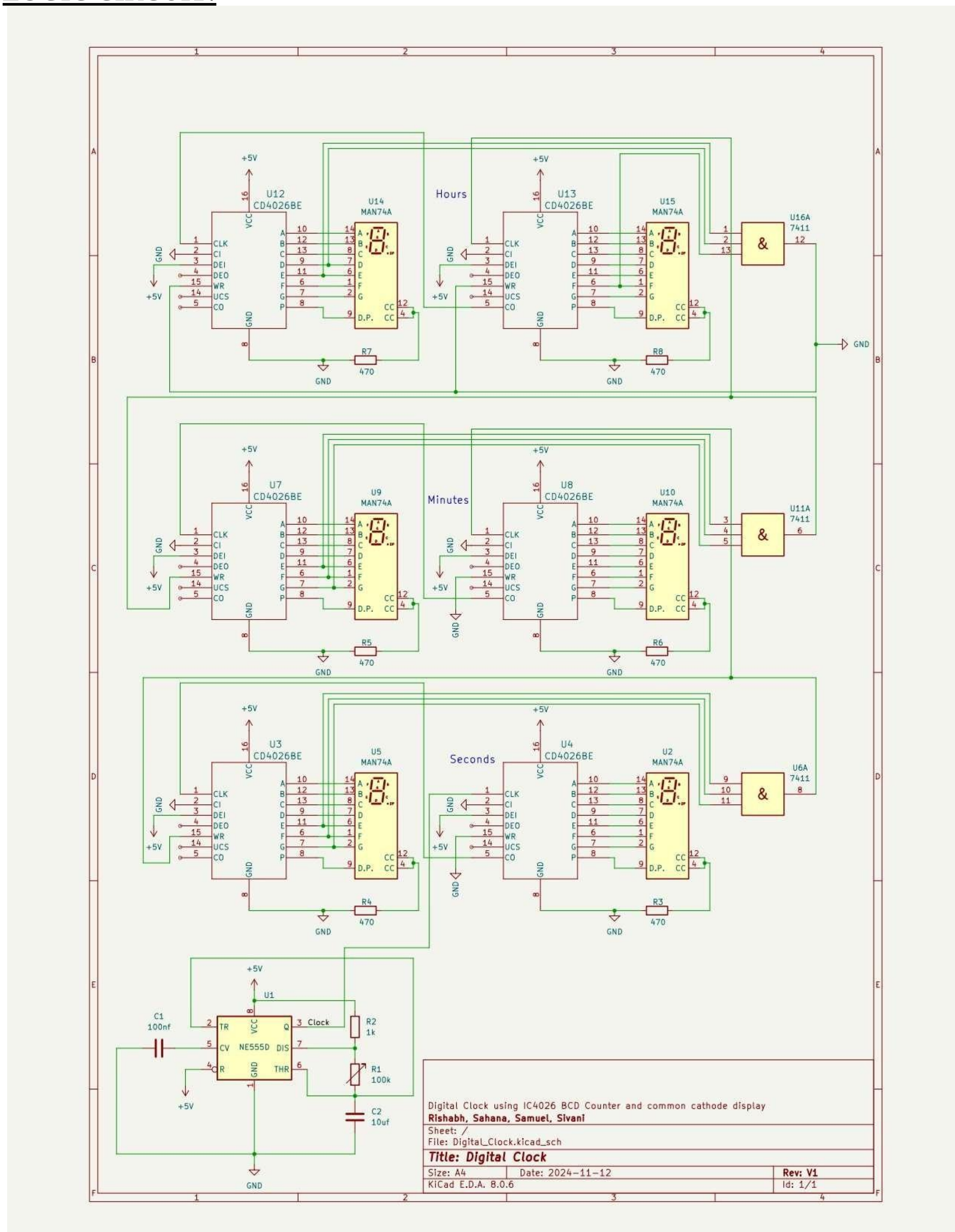
555 Timer IC:
It is used to generate a steady clock pulse. Each pulse increments the seconds by 1.

7 Segment Common Cathode Display:
In a digital clock circuit with 4026 counters, a common cathode 7-segment display is connected so that the common cathode pin goes to ground. Each segment (a–g) of the display is controlled by the 4026 outputs, which light up the necessary segments to show digits 0–9. The 4026 advances the displayed digit with each clock pulse, enabling real-time counting for seconds, minutes, and hours.

Resistors and Capacitors:
In a digital clock, resistors and capacitors are used to set precise timing intervals, such as generating a 1 Hz pulse for counting seconds. They also smooth input signals (debouncing) and limit current to 7-segment displays to protect components. Capacitors help stabilize voltage and prevent fluctuations in the timing circuit.

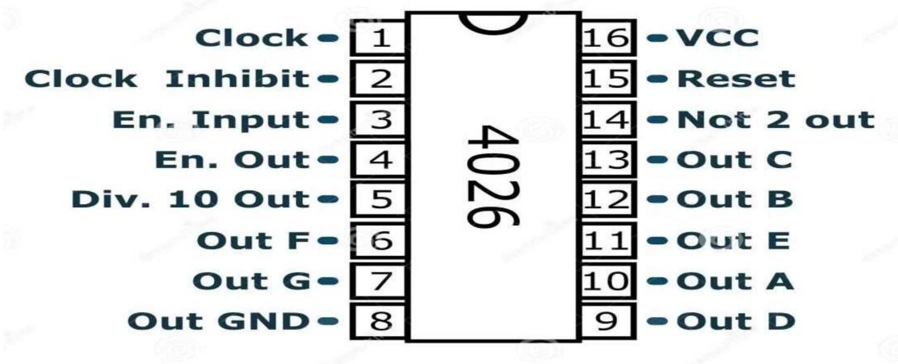
100K Potentiometer:
The 100k potentiometer connected to a timer IC 555 is used to adjust the timing interval. By varying the resistance, it changes the charge/discharge rate of the capacitor, altering the output pulse frequency. This allows fine-tuning of the clock's timing accuracy or pulse duration.

LOGIC CIRCUIT:

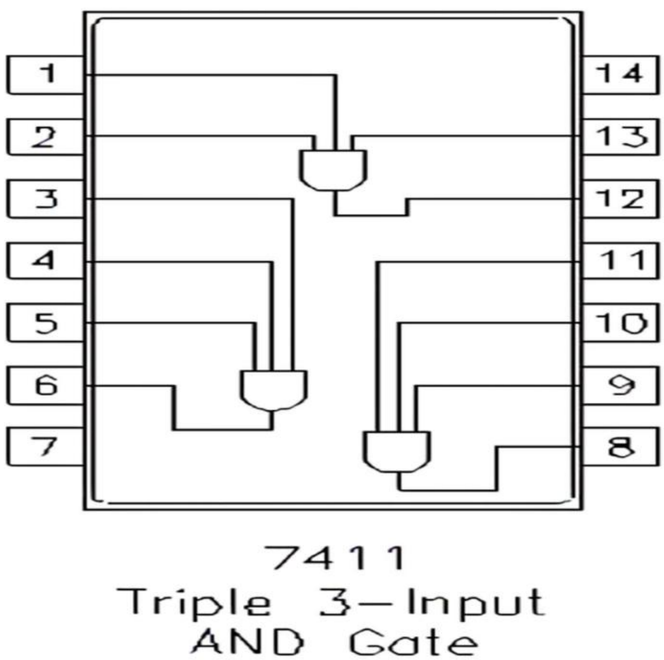


BLOCK DIAGRAM CONSISTING OF ALL THE REQUIRED FUNCTIONS:

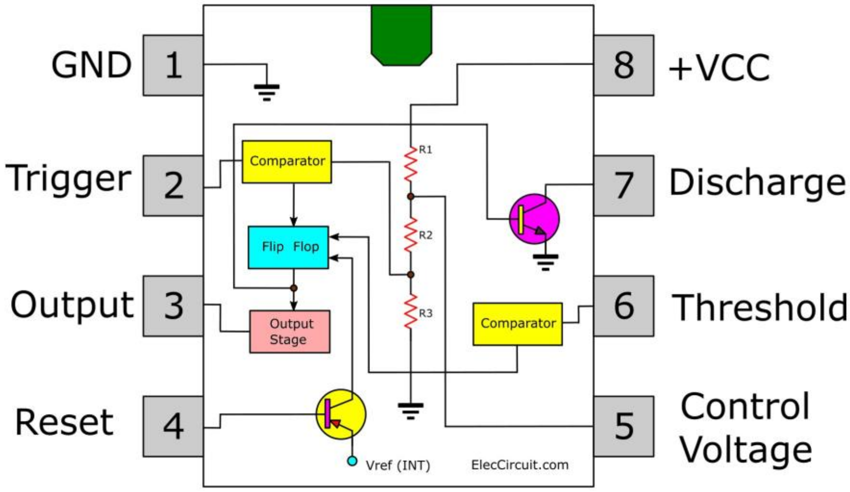
Pin Configuration of IC 4026:



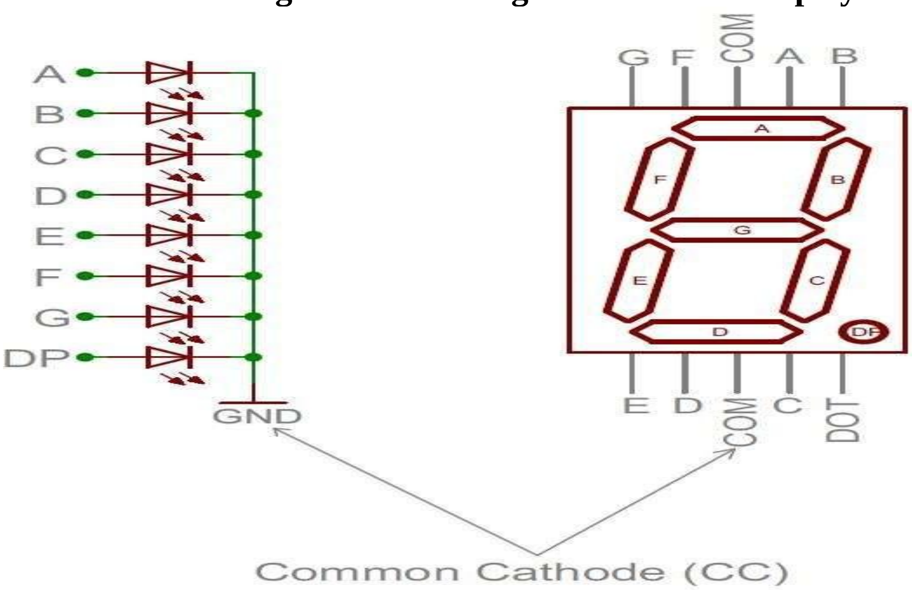
Pin Configuration of AND Gate 7411:



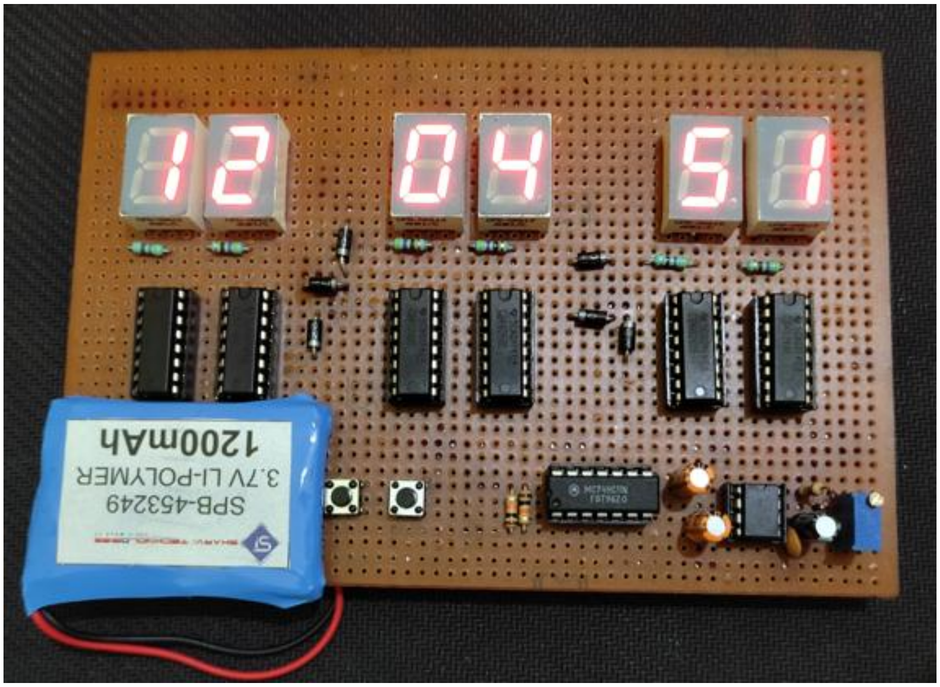
Pin Configuration and Block Diagram of 555 IC:



Pin Configuration of 7 Segment Cathode Display:



SCREENSHOT OF THE WORKING HARDWARE FOR ALL THE REQUIRED FUNCTIONS:



RESULT:

The digital clock successfully displayed time in HH:MM:SS format on the common cathode 7-segment display. IC 4026 stored and shifted time data, while IC 7411 managed the counting logic. The 555 IC ensured accurate timing with a steady 1KHz pulse.

CONCLUSION:

This experiment successfully demonstrated the construction of a digital clock using IC 4076 and IC 7411and a 7-segment common cathode display. The clock displayed time in HH:MM:SS format accurately, with stable timing provided by the 555 IC.

PROBLEMS FACED DURING THE IMPLEMENTATION:

- Problems were faced with Implementation of 60 mod counting for seconds and minutes. The counts were going up till 99 and then to 00 instead of going up to 59 to 00.
- Problems were faced with the implementation of hours segment of the digital clock. The hours segment wouldn't increase the count when the minutes segment had crossed 59 and reset.

- Invalid Outputs were obtained due to logic errors caused in the AND gates which was due to a high voltage input supply.
- Formation of solder bridges (short circuit) and cold joints.
- Was unable to set hours using button as we did for minutes – (the RC circuit was not working).

PROBLEMS RESOLVED USING SOLUTIONS:

After analyzing the circuit and trying to debug the issues, the problems mentioned have been resolved.

- 60 mod counting issue was due to a high frequency clock pulse set through the timer IC555. A dysfunctional AND gate was also suspected to be the underlying issue.
- Issue with the AND gate was causing the segment of hours display to not update count even after minutes reset. Issue was resolved by using another 3-input-AND-Gate 7411IC.
- A slightly higher power supply was causing logic errors in the ICs and therefore resulting in invalid outputs. The issue was fixed by lowering the input voltage and keeping the circuit off for a small duration.
- Ensure the soldering iron temperature is appropriate and the solder is applied properly.